

MATH 3551 A — Abstract Algebra I

Additional Practice Midterm 1 — Version B — SOLUTIONS

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About Version B. This additional practice exam was generated with GPT-5.6 Terra in Codex. It targets about thirty minutes of work in this fifty-minute period. Please report any error, ambiguity, out-of-coverage item, or unexpectedly time-consuming question to Professor Dupuy.

Problem 1. (30 points) **True or false; prove or disprove.**

- (a) **TRUE.** For example, $x = 2$ works because $14 \cdot 2 = 28 \equiv 8 \pmod{20}$. Equivalently, $\gcd(14, 20) = 2$ divides 8.
- (b) **FALSE.** In $(\mathbb{Z}, +)$, let $H = 2\mathbb{Z}$ and $K = 3\mathbb{Z}$. Both are subgroups, but $2 \in H$ and $3 \in K$ while $2 + 3 = 5 \notin H \cup K$.
- (c) **FALSE.** If r is a rotation and s is a reflection, then two reflections can be written sr^i and sr^j , and $(sr^i)(sr^j) = r^{j-i}$, a rotation. For instance, $s(sr) = r$.
- (d) **TRUE.** For $X, Y \in \mathrm{GL}_2(\mathbb{R})$,

$$\det(XY) = \det(X) \det(Y),$$

which is exactly the homomorphism condition into the multiplicative group $\mathbb{R}^\times$.

- (e) **TRUE.** The element $(\bar{1}, \bar{1})$ has order $\mathrm{lcm}(2, 3) = 6$, which is the order of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Hence it generates the whole group.
- (f) **FALSE.** The trivial homomorphism $\varphi: G \rightarrow H$ sends every element of G to 1_H . If G has a nonidentity element g , then $\varphi(g) = 1_H$ although $g \neq 1_G$.

Grading note. Award two points for the circle and three for a specific reason. For (b), any two subgroups whose union fails closure are sufficient. For (c), the key distinction is that a product of two reflections is a rotation. In (f), a valid counterexample to injectivity earns full reason-credit; students who state instead that the conclusion holds for an <i>injective</i> homomorphism have identified the missing hypothesis and earn the reason-credit.

Problem 2. (25 points) **A reduction homomorphism.**

(a) Suppose $\bar{a} = \bar{b}$ in $\mathbb{Z}/24\mathbb{Z}$. Then $24 \mid (a - b)$, so certainly $8 \mid (a - b)$. Thus $\bar{a} = \bar{b}$ in $\mathbb{Z}/8\mathbb{Z}$, and π is well defined.

(b) For all $\bar{a}, \bar{b} \in \mathbb{Z}/24\mathbb{Z}$,

$$\pi(\bar{a} + \bar{b}) = \pi(\overline{a + b}) = \overline{a + b} = \bar{a} + \bar{b} = \pi(\bar{a}) + \pi(\bar{b}).$$

So π is a homomorphism.

(c) The kernel consists of those classes divisible by 8:

$$\ker \pi = \{\bar{0}, \bar{8}, \bar{16}\} = \langle \bar{8} \rangle \leq \mathbb{Z}/24\mathbb{Z}.$$

Every class in $\mathbb{Z}/8\mathbb{Z}$ is $\pi(\bar{a})$ for the same representative a , so $\text{im } \pi = \mathbb{Z}/8\mathbb{Z}$.

(d) In $\mathbb{Z}/24\mathbb{Z}$,

$$K = \langle \bar{6} \rangle = \{\bar{0}, \bar{6}, \bar{12}, \bar{18}\}.$$

Therefore

$$\pi(K) = \{\bar{0}, \bar{6}, \bar{4}, \bar{2}\} = \langle \bar{2} \rangle \leq \mathbb{Z}/8\mathbb{Z}.$$

(e) A class $\bar{a}$ maps to a generator of $\mathbb{Z}/8\mathbb{Z}$ exactly when $\gcd(a, 8) = 1$, i.e. exactly when a is odd. Thus the twelve desired classes in $\mathbb{Z}/24\mathbb{Z}$ are

$$\bar{1}, \bar{3}, \bar{5}, \bar{7}, \bar{9}, \bar{11}, \bar{13}, \bar{15}, \bar{17}, \bar{19}, \bar{21}, \bar{23}.$$

Grading note. In (a), the implication $24 \mid (a - b) \Rightarrow 8 \mid (a - b)$ is the whole well-definedness proof. In (c), award three points for the correct kernel and two for the surjective image. In (e), award four points for the criterion $\gcd(a, 8) = 1$ (or “ a odd”) and four for all twelve classes; do not give full credit for merely listing the four generators of $\mathbb{Z}/8\mathbb{Z}$.

Problem 3. (25 points) **A matrix group over $\mathbb{F}_2$.**

(a) Direct multiplication gives

$$A^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad A^3 = A^2 A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Since $A \neq I$ and $A^2 \neq I$, the order of A is 3.

(b) Direct multiplication gives

$$BAB = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = A^2 = A^{-1}.$$

(c) The six matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \\ B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad AB = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad A^2 B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

all lie in H and are visibly distinct. A matrix in $\text{GL}_2(\mathbb{F}_2)$ is determined by its two linearly independent columns. There are 3 choices for the first nonzero column and then 2 choices for a second column not equal to it, hence

$$|\text{GL}_2(\mathbb{F}_2)| = 3 \cdot 2 = 6.$$

Thus H , which already contains six distinct elements of $\text{GL}_2(\mathbb{F}_2)$, is all of $\text{GL}_2(\mathbb{F}_2)$.

(d)

$$AB = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad (AB)^2 = I.$$

Since $AB \neq I$, it has order 2.

Grading note. In (a) and (d), both a correct product and the conclusion about the order are required. In (b), the intended observation is the relation $BAB = A^{-1}$. For (c), award six points for the six correct, distinct matrices and four for the count-and-containment proof. A student may count invertible matrices by determinants instead and receive full credit.

Problem 4. (20 points) **A subgroup of $\mathbb{Z} \times \mathbb{Z}$.**

- (a) The identity $(0, 0)$ belongs to H . If $(a, b), (c, d) \in H$, then $3 \mid (a + b)$ and $3 \mid (c + d)$, so

$$3 \mid (a + b) - (c + d) = (a - c) + (b - d).$$

Hence $(a, b) - (c, d) \in H$. The subgroup test shows that $H \leq G$.

- (b) Both $(1, -1)$ and $(0, 3)$ lie in H , and H is a subgroup by (a), so

$$\langle (1, -1), (0, 3) \rangle \subseteq H.$$

Conversely, let $(a, b) \in H$. Write $a + b = 3q$ for some $q \in \mathbb{Z}$. Then

$$(a, b) = a(1, -1) + q(0, 3),$$

because the second coordinate is $-a + 3q = -a + (a + b) = b$. Thus every element of H is in the generated subgroup, proving equality.

- (c) Suppose, for contradiction, that $H = \langle (a, b) \rangle$. Since $(1, -1) \in H$, we have $(1, -1) = m(a, b)$ for some $m \in \mathbb{Z}$. The first coordinate says $1 = ma$, so $m = \pm 1$ and $(a, b) = \pm(1, -1)$. But every multiple of $(1, -1)$ has coordinate sum 0, whereas $(0, 3) \in H$ has coordinate sum 3. This contradiction proves that H is not cyclic.

Grading note. For (a), students may use the one-step subgroup test; closure under subtraction must be shown, not only asserted. In (b), give four points for the easy inclusion and five for writing an arbitrary (a, b) with $a + b = 3q$ as $a(1, -1) + q(0, 3)$. In (c), the essential proof is that any proposed single generator is forced by $(1, -1)$, then cannot generate $(0, 3)$; a bare claim that “there are two generators” does not suffice.